Name: _____
ID: _____

MTH234 Linear Algebra
1/2025 (Regular/International Section)
10th September 2025
3 hrs. (09:00 - 12:00 AM.)

MODULE 1 MOCK EXAM (Total 50 points)

Instruction: Students must provide all details of your work. You may use any class materials. Calculators and other materials are not allowed.

(1) (6 points) Solve the matrix equation:

$$\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} X^T - \begin{bmatrix} 1 & 4 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ 5 & 4 \end{bmatrix} \quad \text{for } X.$$

where T denotes the transpose of a matrix.

Sol!! จากโจทย์ จะได้ว่า

$$\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} X^T = \begin{bmatrix} 2 & -2 \\ 5 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 4 \\ 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} X^T = \begin{bmatrix} 3 & 2 \\ 7 & 4 \end{bmatrix}$$

$$\underbrace{\frac{1}{7}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}}_{A^{-1}} \underbrace{\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}}_{A} X^T = \frac{1}{7}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 3 & 2 \\ 7 & 4 \end{bmatrix}$$

จาก $A^{-1}A = I$;

จาก $IA = A$;

$$IX^T = \frac{1}{7}\begin{bmatrix} 6-7 & 4-4 \\ 3+21 & 2+12 \end{bmatrix}$$

$$X^T = \frac{1}{7}\begin{bmatrix} -1 & 0 \\ 24 & 14 \end{bmatrix}$$

$$X^T = \begin{bmatrix} -\frac{1}{7} & -\frac{0}{7} \\ \frac{24}{7} & \frac{14}{7} \end{bmatrix} = \begin{bmatrix} -\frac{1}{7} & 0 \\ \frac{24}{7} & 2 \end{bmatrix}$$

$\therefore$ $$X = \begin{bmatrix} -\frac{1}{7} & \frac{24}{7} \\ 0 & 2 \end{bmatrix}$$ // Ans.

พิจารณาว่าให้ $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$

จะได้ว่า $A^{-1} = \frac{1}{\det A}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$

$= \frac{1}{6+1}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$

$= \frac{1}{7}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$

(2) (10 points) Use elementary row operations to find $\det(A)$ and A^{-1} (if any) where

$$A = \begin{bmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 3 & 0 \\ 2 & 1 & -5 & 3 \end{bmatrix}$$

Sol: จากโจทย์ จะได้ว่า

$$\begin{bmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 3 & 0 \\ 2 & 1 & -5 & 3 \end{bmatrix} \sim \begin{bmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 3 & 0 \\ 0 & 1 & -5 & 1 \end{bmatrix} R_4 - 2R_2$$

$$\sim \begin{bmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & -2 & 1 \end{bmatrix} R_4 + R_3$$

$$\sim \begin{bmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} R_4 + R_1$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} R_3 \leftrightarrow R_2$$

จากเมทริกซ์ที่ได้เป็นเมทริกซ์สามเหลี่ยมบน

ทำให้ได้ว่า $\det A = (1)(-1)(2)(1)$

$\det A = -2$ //

จะเห็นว่า $\det A \neq 0$ จึงสามารถหา A^{-1} ได้

ใช้หลักการดำเนินการด้วยแถว โดยแปลงจากเมทริกซ์เอกฐาน $[A|I] \sim [I|A^{-1}]$

จะได้ว่า

$$\left[\begin{array}{cccc|cccc} 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 3 & 0 & 0 & 0 & 1 & 0 \\ 2 & 1 & -5 & 3 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{cccc|cccc} 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 & 0 & -1 & 0 \\ 2 & 1 & -5 & 3 & 0 & 0 & 0 & 1 \end{array}\right] -R_3$$

$$\sim \left[\begin{array}{cccc|cccc} 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & -5 & 1 & 0 & -2 & 0 & 1 \end{array}\right] R_4 - 2R_3$$

$$\sim \left[\begin{array}{cccc|cccc} 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & -2 & 1 & 1 \end{array}\right] R_4 - R_3 + R_1$$

$$\sim \left[\begin{array}{cccc|cccc} 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & \frac{3}{2} & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & -2 & 1 & 1 \end{array}\right] R_3 + \frac{3}{2}R_1$$

$$\sim \left[\begin{array}{cccc|cccc} 0 & 0 & 1 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & 3 & -1 & -1 \\ 0 & 1 & 0 & 0 & \frac{3}{2} & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & -2 & 1 & 1 \end{array}\right] \begin{array}{l} \frac{1}{2}R_1 \\ R_2 - R_4 \end{array}$$

$$\sim \left[\begin{array}{cccc|cccc} 0 & 1 & 0 & 0 & \frac{3}{2} & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & -1 & 3 & -1 & -1 \\ 0 & 0 & 1 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & -2 & 1 & 1 \end{array}\right] R_3 \leftrightarrow R_2$$

$$\sim \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -1 & 3 & -1 & -1 \\ 0 & 1 & 0 & 0 & \frac{3}{2} & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & -2 & 1 & 1 \end{array}\right] R_2 \leftrightarrow R_1$$

$\therefore$ A^{-1} ที่ได้คือ $\begin{bmatrix} -1 & 3 & -1 & -1 \\ \frac{3}{2} & 0 & -1 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \\ 1 & -2 & 1 & 1 \end{bmatrix}$ // Ans.

(3) (8 points) Use Cramer's rule to solve the following system of equations:

$$\begin{aligned} 2x_1 + x_2 - x_3 &= 4 \\ x_1 - 2x_2 + x_3 &= 1 \\ 3x_1 - 3x_2 + 2x_3 &= 0 \end{aligned}$$

Sol: จากโจทย์ จะได้ว่า $AX = \vec{b} \rightarrow \begin{bmatrix} 2 & 1 & -1 \\ 1 & -2 & 1 \\ 3 & -3 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$

จะได้ว่า $\det A = \begin{vmatrix} 2 & 1 & -1 \\ 1 & -2 & 1 \\ 3 & -3 & 2 \end{vmatrix} = -8+3+3-6+6-2 = -4$

$\det A_1 = \begin{vmatrix} 4 & 1 & -1 \\ 1 & -2 & 1 \\ 0 & -3 & 2 \end{vmatrix} = -16+0+3-0+12-2 = -3$

$\det A_2 = \begin{vmatrix} 2 & 4 & -1 \\ 1 & 1 & 1 \\ 3 & 0 & 2 \end{vmatrix} = 4+12+0+3+0-8 = 11$

$\det A_3 = \begin{vmatrix} 2 & 1 & 4 \\ 1 & -2 & 1 \\ 3 & -3 & 0 \end{vmatrix} = 0+3-12+24+6+0 = 21$

ทำให้ได้ว่า $x_1 = \dfrac{\det A_1}{\det A} = \dfrac{-3}{-4} = \dfrac{3}{4}$

$x_2 = \dfrac{\det A_2}{\det A} = \dfrac{11}{-4} = -\dfrac{11}{4}$

$x_3 = \dfrac{\det A_3}{\det A} = \dfrac{21}{-4} = -\dfrac{21}{4}$

$\therefore$ คำตอบของระบบสมการที่ได้คือ $x_1 = \frac{3}{4}, x_2 = -\frac{11}{4}, x_3 = -\frac{21}{4}$ // Ans.

(4) (8 points) Use elementary row operations to determine whether the following system of linear equations is consistent or not. If it is, find the general solution:

$$\begin{aligned} x_1 - 3x_2 + x_3 &= 2 \\ 3x_1 - 8x_2 + 2x_3 &= 5 \\ 2x_2 - 2x_3 &= -2 \end{aligned}$$

Sol: จากโจทย์ จะได้ว่า $[A|b] = \left[\begin{array}{ccc|c} 1 & -3 & 1 & 2 \\ 3 & -8 & 2 & 5 \\ 0 & 2 & -2 & -2 \end{array}\right]$

$$\sim \left[\begin{array}{ccc|c} 1 & -3 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 2 & -2 & -2 \end{array}\right] \quad R_2 - 3R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & -3 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 3 & -1 & -3 \end{array}\right] \quad R_3 + R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 4 & 0 & -4 \\ 0 & 3 & -1 & -3 \end{array}\right] \quad \begin{array}{l} R_1 + R_3 \\ R_2 + R_3 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 3 & -1 & -3 \end{array}\right] \quad \frac{1}{4}R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{array}\right] \quad R_3 - 3R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

∴ คำตอบของระบบสมการที่ได้คือ $x_1 = -1, x_2 = -1, x_3 = 0$ // Ans.

(5) (8 points) Find all values of a that make the following linear system:

$$\begin{aligned} x + y - z &= 1 \\ 2x + 3y + az &= 3 \\ x + ay + 3z &= 2 \end{aligned}$$

fall into each of the following cases:

(a) No solution

(b) A unique solution

(c) Infinitely many solutions

Sol: จากโจทย์ จะได้ว่า $[A|\vec{b}] = \left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 2 & 3 & a & 3 \\ 1 & a & 3 & 2 \end{array}\right]$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & a+2 & 1 \\ 0 & a-1 & 4 & 1 \end{array}\right] \begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & a+2 & 1 \\ 0 & 0 & (3+a)(2-a) & 2-a \end{array}\right] R_3 - (a-1)R_2$$

a) ไม่มีผลเฉลย เมื่อ $(3+a)(2-a) = 0$ และ $2-a \neq 0$

$a = -3, \cancel{2}$

$\therefore$ $a = 3$ # Ans.

b) มีผลเฉลยเพียงคำตอบเดียว เมื่อ $(3+a)(2-a) \neq 0$

$\therefore$ $a \neq -3, 2$ # Ans.

c) มีผลเฉลยคำตอบเป็นอนันต์ เมื่อ $(3+a)(2-a) = 0$ และ $2-a = 0$

$a = \cancel{-3}, 2$

$\therefore$ $a = 2$ # Ans.

(6) (10 points) For the following system of linear equations:

$$\begin{aligned} w + 2x + 2y + 2z &= 1 \\ w + 3x + 4y + 3z &= 6 \\ 2w + 6x + 9y + 8z &= 13 \\ w + 5x + 8y + \alpha z &= \beta \end{aligned}$$

Find all values of α and β that make the system have a solution in the following cases:

(a) Case $\alpha = 6,\ \beta = 17$

(b) Case $\alpha = 5,\ \beta = 16$

Sol: จากโจทย์ จะได้ว่า

1.)
$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & 1 \\ 1 & 3 & 4 & 3 & 6 \\ 2 & 6 & 9 & 8 & 13 \\ 1 & 5 & 8 & 6 & 17 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & 1 \\ 0 & 1 & 2 & 1 & 5 \\ 0 & 2 & 5 & 4 & 11 \\ 0 & 3 & 6 & 4 & 16 \end{array}\right] \begin{array}{l} R_2 - R_1 \\ R_3 - 2R_1 \\ R_4 - R_1 \end{array} \sim \left[\begin{array}{cccc|c} 1 & 0 & -2 & 0 & -9 \\ 0 & 1 & 2 & 1 & 5 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right] \begin{array}{l} R_1 - 2R_2 \\ \\ R_3 - 2R_2 \\ R_4 - 3R_2 \end{array}$$

ทำให้ได้ว่า
$$\begin{aligned} w - 2y &= -9 \\ x + 2y + z &= 5 \\ y + 2z &= 1 \\ z &= 1 \end{aligned}$$

จะได้ $y = 1 - 2(1) = -1$

$x = 5 - 1 - 2(-1) = 6$

$w = -9 + 2(-1) = -11$

$\therefore w = -11, x = 6, y = -1, z = 1$ # Ans.

2)
$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & 1 \\ 1 & 3 & 4 & 3 & 6 \\ 2 & 6 & 9 & 8 & 13 \\ 1 & 5 & 8 & 5 & 16 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & 1 \\ 0 & 1 & 2 & 1 & 5 \\ 0 & 2 & 5 & 4 & 11 \\ 0 & 3 & 6 & 3 & 15 \end{array}\right] \begin{array}{l} R_2 - R_1 \\ R_3 - 2R_1 \\ R_4 - R_1 \end{array} \sim \left[\begin{array}{cccc|c} 1 & 0 & -2 & 0 & -9 \\ 0 & 1 & 2 & 1 & 5 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} R_1 - 2R_2 \\ \\ R_3 - 2R_2 \\ R_4 - 3R_2 \end{array}$$

ทำให้ได้ว่า
$$\begin{aligned} w - 2y &= -9 \\ x + 2y + z &= 5 \\ y + 2z &= 1 \end{aligned}$$

โดยที่ $z = t \in \mathbb{R}$

ทำให้ได้ว่า $y = -2t + 1$

$x = 5 - t - 2(1 - 2t) = 5 - t - 2 + 4t = 3t + 3$

$w = -9 + 2(1 - 2t) = -9 + 2 - 4t = -4t - 7$

$\therefore \vec{v} = t\begin{bmatrix} -4 \\ 3 \\ -2 \\ 1 \end{bmatrix} + \begin{bmatrix} -7 \\ 3 \\ 1 \\ 0 \end{bmatrix}$ # Ans.